DLMI with Berstain Polynomial

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1 Inspiration

Let's focus on one interval $[\rho^k, \rho^{k+1}]$. Define the normalized variable

$$\alpha = \frac{\rho^{k+1} - \rho}{\rho^{k+1} - \rho^k} \in [0, 1],$$

and consider the following multiplication:

$$\begin{aligned} P(\alpha)A(\alpha) &= [\alpha P_0 + (1-\alpha)P_1]\,[\alpha A_0 + (1-\alpha)A_1] \\ &= \alpha^2 P_0 A_0 + \alpha(1-\alpha)(P_0 A_1 + P_1 A_0) + (1-\alpha)^2 P_1 A_1. \end{aligned} \tag{1}$$

1.1 Bernstein Basis Polynomial

Definition 1 (Bernstein basis polynomial): For a nonnegative integer n, the Bernstein basis polynomials of degree n on $[0, 1]$ are defined as

$$B_{i,n}(\alpha) = \binom{n}{i} \alpha^{n-i} (1-\alpha)^i, \quad i = 0, 1, \ldots, n.$$

Example 1: The Bernstein basis polynomials of degree $0, 1, 2, 3$ are listed as follows:

$$\begin{aligned} n = 0 :&\quad B_{0,0}(\alpha) = 1, \\ n = 1 :&\quad B_{0,1}(\alpha) = \alpha, \quad B_{1,1}(\alpha) = 1 - \alpha, \\ n = 2 :&\quad B_{0,2}(\alpha) = \alpha^2, \quad B_{1,2}(\alpha) = 2\alpha(1-\alpha), \quad B_{2,2}(\alpha) = (1-\alpha)^2, \\ n = 3 :&\quad B_{0,3}(\alpha) = \alpha^3, \quad B_{1,3}(\alpha) = 3\alpha^2(1-\alpha), \\ &\quad B_{2,3}(\alpha) = 3\alpha(1-\alpha)^2, \quad B_{3,3}(\alpha) = (1-\alpha)^3. \end{aligned}$$

(1) shows that the product of two linear splines is a quadratic polynomial in α. Equivalently, this product can be expressed by the degree-2 Bernstein basis polynomials as

$$P(\alpha)A(\alpha) = B_{0,2}(\alpha) P_0 A_0 + \frac{1}{2} B_{1,2}(\alpha)(P_0 A_1 + P_1 A_0) + B_{2,2}(\alpha) P_1 A_1.$$

If we focus on the red terms, they are exactly the convolution of the two linear coefficient sequences $[P_0, P_1]$ and $[A_0, A_1]$, namely

$$[P_0, P_1] * [A_0, A_1] = [P_0 A_0,\ P_0 A_1 + P_1 A_0,\ P_1 A_1].$$

The factor $1/2$ in the middle Bernstein coefficient comes from the binomial coefficient in $B_{1,2}(\alpha) = 2\alpha(1-\alpha)$. More generally, using the Bernstein basis polynomial representation, we can equivalently express the multiplication of m linear splines by the degree-m Bernstein basis, which contains $m + 1$ basis polynomials. Let

$$S_r(\alpha) = \alpha S_{r,0} + (1-\alpha) S_{r,1}, \quad r = 1, \ldots, m.$$

Then

$$\prod_{r=1}^{m} S_r(\alpha) = \sum_{i=0}^{m} B_{i,m}(\alpha) C_i,$$

where

$$C_i = \frac{1}{\binom{m}{i}} \sum_{\substack{\nu_1 + \cdots + \nu_m = i \\ \nu_r \in \{0,1\}}} S_{1,\nu_1} S_{2,\nu_2} \cdots S_{m,\nu_m}, \quad i = 0, 1, \ldots, m.$$

Equivalently, if

$$[D_0, D_1, \ldots, D_m] = [S_{1,0}, S_{1,1}] * [S_{2,0}, S_{2,1}] * \cdots * [S_{m,0}, S_{m,1}],$$

then the Bernstein coefficients satisfy

$$C_i = \frac{D_i}{\binom{m}{i}}, \quad i = 0, 1, \ldots, m.$$

Similarly, the product of a degree-n Bernstein expression and a degree-m Bernstein expression is a degree-$(n+m)$ Bernstein expression. Specifically, if

$$P(\alpha) = \sum_{i=0}^{n} P_i B_{i,n}(\alpha), \quad Q(\alpha) = \sum_{j=0}^{m} Q_j B_{j,m}(\alpha),$$

then

$$P(\alpha)Q(\alpha) = \sum_{k=0}^{n+m} \frac{1}{\binom{n+m}{k}} B_{k,n+m}(\alpha) D_k, \quad \mathbf{D}^{(n+m)} = [D_0, \ldots, D_{n+m}].$$

Define the weighted coefficient vectors

$$\mathbf{P}^{(n)} = \left[\binom{n}{0} P_0, \ldots, \binom{n}{n} P_n\right], \quad \mathbf{Q}^{(m)} = \left[\binom{m}{0} Q_0, \ldots, \binom{m}{m} Q_m\right].$$

Then the coefficient vector in the above expression is simply

$$\mathbf{D}^{(n+m)} = \mathbf{P}^{(n)} * \mathbf{Q}^{(m)}.$$

Here $*$ denotes discrete convolution. In this form, the binomial normalization is multiplied back to the Bernstein basis term, just as the factor $\frac{1}{2}$ is placed in front of $B_{1,2}(\alpha)$ in the quadratic case.

1.2 Multi-variable case

The above construction is not limited to a scalar parameter α. Suppose the scheduling parameter is multi-dimensional, i.e., $\rho \in \mathbb{R}^\ell$. On a hyper-rectangle $[\rho^k, \rho^{k+1}]$, define the normalized variables

$$\alpha_j = \frac{\rho_j^{k+1} - \rho_j}{\rho_j^{k+1} - \rho_j^k} \in [0, 1], \quad j = 1, \ldots, \ell.$$

Then a linear spline in each parameter direction becomes a multilinear model.

For a multi-index $\mathbf{n} = (n_1, \ldots, n_\ell)$ and $\mathbf{i} = (i_1, \ldots, i_\ell)$ with $0 \le i_j \le n_j$, the multi-variable Bernstein basis polynomial is defined as the tensor product

$$B_{\mathbf{i},\mathbf{n}}(\boldsymbol{\alpha}) = \prod_{j=1}^{\ell} B_{i_j,n_j}(\alpha_j) = \prod_{j=1}^{\ell} \binom{n_j}{i_j} \alpha_j^{n_j - i_j} (1 - \alpha_j)^{i_j}.$$

Example 2 (Two-dimensional multilinear basis): For $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)$ and $\mathbf{1}_2 = (1, 1)$, the Bernstein basis polynomials are

$$\begin{aligned}
B_{(0,0),\mathbf{1}_2}(\boldsymbol{\alpha}) &= \alpha_1 \alpha_2, \\
B_{(1,0),\mathbf{1}_2}(\boldsymbol{\alpha}) &= (1 - \alpha_1)\alpha_2, \\
B_{(0,1),\mathbf{1}_2}(\boldsymbol{\alpha}) &= \alpha_1(1 - \alpha_2), \\
B_{(1,1),\mathbf{1}_2}(\boldsymbol{\alpha}) &= (1 - \alpha_1)(1 - \alpha_2).
\end{aligned}$$

Hence a two-dimensional multilinear model can be written as

$$S(\boldsymbol{\alpha}) = \alpha_1 \alpha_2 S_{0,0} + (1 - \alpha_1)\alpha_2 S_{1,0} + \alpha_1(1 - \alpha_2) S_{0,1} + (1 - \alpha_1)(1 - \alpha_2) S_{1,1}.$$

Now consider two two-dimensional multilinear models:

$$\begin{aligned}
P(\alpha_1, \alpha_2) &= \alpha_1 \alpha_2 P_{0,0} + (1 - \alpha_1)\alpha_2 P_{1,0} + \alpha_1(1 - \alpha_2) P_{0,1} + (1 - \alpha_1)(1 - \alpha_2) P_{1,1}, \\
Q(\alpha_1, \alpha_2) &= \alpha_1 \alpha_2 Q_{0,0} + (1 - \alpha_1)\alpha_2 Q_{1,0} + \alpha_1(1 - \alpha_2) Q_{0,1} + (1 - \alpha_1)(1 - \alpha_2) Q_{1,1}.
\end{aligned}$$

Their product is a Bernstein polynomial with degree multi-index $\mathbf{2}_2 = (2, 2)$:

$$\begin{aligned}P(\alpha_1, \alpha_2)Q(\alpha_1, \alpha_2) = {} & \alpha_1^2\alpha_2^2 D_{0,0} + \alpha_1(1-\alpha_1)\alpha_2^2 D_{1,0} + (1-\alpha_1)^2\alpha_2^2 D_{2,0} \\ & + \alpha_1^2\alpha_2(1-\alpha_2)D_{0,1} + \alpha_1(1-\alpha_1)\alpha_2(1-\alpha_2)D_{1,1} + (1-\alpha_1)^2\alpha_2(1-\alpha_2)D_{2,1} \\ & + \alpha_1^2(1-\alpha_2)^2 D_{0,2} + \alpha_1(1-\alpha_1)(1-\alpha_2)^2 D_{1,2} + (1-\alpha_1)^2(1-\alpha_2)^2 D_{2,2}.\end{aligned}$$

Here the coefficients $D_{p,q}$ are given by the two-dimensional convolution of the two 2×2 coefficient arrays:

$$D = \begin{bmatrix} D_{0,0} & D_{0,1} & D_{0,2} \\ D_{1,0} & D_{1,1} & D_{1,2} \\ D_{2,0} & D_{2,1} & D_{2,2} \end{bmatrix} = \begin{bmatrix} P_{0,0} & P_{0,1} \\ P_{1,0} & P_{1,1} \end{bmatrix} *_{\text{2-Dim}} \begin{bmatrix} Q_{0,0} & Q_{0,1} \\ Q_{1,0} & Q_{1,1} \end{bmatrix}.$$

Equivalently,

$$\begin{aligned} &D_{0,0} = P_{0,0}Q_{0,0}, && D_{1,0} = P_{1,0}Q_{0,0} + P_{0,0}Q_{1,0}, && D_{2,0} = P_{1,0}Q_{1,0}, \\ &D_{0,1} = P_{0,1}Q_{0,0} + P_{0,0}Q_{0,1}, && D_{1,1} = P_{1,1}Q_{0,0} + P_{1,0}Q_{0,1} + P_{0,1}Q_{1,0} + P_{0,0}Q_{1,1}, && D_{2,1} = P_{1,1}Q_{1,0} + P_{1,0}Q_{1,1}, \\ &D_{0,2} = P_{0,1}Q_{0,1}, && D_{1,2} = P_{1,1}Q_{0,1} + P_{0,1}Q_{1,1}, && D_{2,2} = P_{1,1}Q_{1,1}. \end{aligned}$$

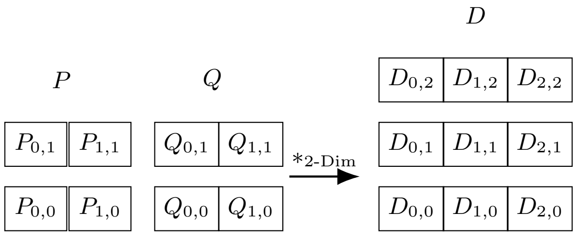

Figure 1: Two-dimensional convolution of multilinear coefficient arrays: two 2×2 arrays generate one 3×3 coefficient array.

The same construction extends to the general ℓ-dimensional parameter case. Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_\ell)$ and $\mathbf{1}_\ell = (1, \ldots, 1)$. For two multilinear models

$$P(\boldsymbol{\alpha}) = \sum_{\boldsymbol{\mu} \in \{0,1\}^\ell} B_{\boldsymbol{\mu}, \mathbf{1}_\ell}(\boldsymbol{\alpha}) P_{\boldsymbol{\mu}}, \quad Q(\boldsymbol{\alpha}) = \sum_{\boldsymbol{\nu} \in \{0,1\}^\ell} B_{\boldsymbol{\nu}, \mathbf{1}_\ell}(\boldsymbol{\alpha}) Q_{\boldsymbol{\nu}},$$

their product has degree multi-index $\mathbf{2}_\ell = (2, \ldots, 2)$ and is given by

$$P(\boldsymbol{\alpha})Q(\boldsymbol{\alpha}) = \sum_{\mathbf{k} \in \{0,1,2\}^\ell} \frac{1}{\prod_{r=1}^{\ell} \binom{2}{k_r}} B_{\mathbf{k}, \mathbf{2}_\ell}(\boldsymbol{\alpha}) D_{\mathbf{k}},$$

where the unscaled convolution tensor is the ℓ-dimensional convolution

$$D_{\mathbf{k}} = \sum_{\boldsymbol{\mu} + \boldsymbol{\nu} = \mathbf{k}} P_{\boldsymbol{\mu}} Q_{\boldsymbol{\nu}} = (P *_{\ell\text{-Dim}} Q)_{\mathbf{k}}.$$

Thus, once the scheduling parameter becomes ℓ-dimensional, the scalar coefficient convolution is replaced by an ℓ-dimensional tensor convolution.

2 Relaxation of the DP-LMI

2.1 The trivial idea

A representitive DP-LMI can be expressed as

$$E(\alpha) \prec 0. \tag{2}$$

where $E(\alpha)$ is a matrix-valued function of the scheduling parameter $\alpha \in \mathbb{R}^\ell$, and $E(\alpha)$ is based on m-degree Bernstein polynomial:

$$E(\alpha) = \sum_{\mathbf{i} \in \{0,\ldots,m\}^\ell} B_{\mathbf{i}, \mathbf{m}_\ell}(\alpha) E_{\mathbf{i}}.$$

Example 3: Still the same case as in (1), for $\ell = 1, m = 2$, $E(\alpha) = He\{P(\alpha)A(\alpha)\}$, then:

$$E(\alpha) = \alpha^2 E_0 + 2\alpha(1-\alpha)E_1 + (1-\alpha)^2 E_2,$$
$$E_0 = He\{P_0 A_0\}, \quad E_1 = \frac{1}{2}He\{P_0 A_1 + P_1 A_0\}, \quad E_2 = He\{P_1 A_1\}.$$

A necessary condition for (2) to hold is to evaluate $\alpha = 0, 1$:

$$E_0 \prec 0, \quad E_2 \prec 0.$$

But for all $\alpha \in [0,1]$, the middle term E_1 is not considered. One sufficient condition is brutally forcing the middle term to be negative definite:

$$E_1 \prec 0.$$

So a trivial idea, for $\ell = 1$, is to evaluate all the Bernstein coefficients E_i and force them to be negative definite:

Lemma 1: When $\ell = 1$, suppose $E(\alpha)$ is a matrix-valued function based in the Bernstein basis:

$$E(\alpha) = \sum_{i=0}^{m} E_i B_{i,m}(\alpha),$$

where $B_{i,m}(\alpha)$ are Bernstein basis polynomials of degree m. A sufficient condition for the DP-LMI (2) to hold for all $\alpha \in [0,1]$ is:

$$E_i \prec 0, \quad i = 0, 1, \ldots, m.$$

Similarly, for the general ℓ-dimensional case, a sufficient condition is to evaluate all the Bernstein coefficients $E_{\mathbf{i}}$ and force them to be negative definite:

Lemma 2: Suppose $E(\boldsymbol{\alpha})$ is a ℓ-dimensional Bernstein spline model with degree multi-index $\mathbf{m}_\ell = (m, \ldots, m)$:

$$E(\boldsymbol{\alpha}) = \sum_{\mathbf{i} \in \{0,\ldots,m\}^\ell} E_{\mathbf{i}} B_{\mathbf{i},\mathbf{m}_\ell}(\boldsymbol{\alpha}),$$

where ℓ is the dimension of the scheduling parameter, m is the Bernstein degree of E in each parameter direction. Thus $E_{\mathbf{i}}$ is indexed by an ℓ-dimensional tensor of size $(m+1)^\ell$. A sufficient condition for the DP-LMI (2) to hold for all $\boldsymbol{\alpha} \in [0,1]^\ell$ is:

$$E_{\mathbf{i}} \prec 0, \quad \mathbf{i} \in \{0, \ldots, m\}^\ell.$$

But as one can see, the number of sufficient conditions grows exponentially with m^ℓ. While only 2^ℓ conditions are necessary, i.e., the vertices of the hyper-rectangle, the rest (middle points like E_1 in Example 3) of the conditions are not necessarily negative definite. So the trivial idea is too conservative. We need a better relaxation idea.

2.2 A relaxation idea

Still, back to Example 3, we can benefit more from the fact that $E_0 \prec 0, E_2 \prec 0$:

Lemma 3: When $\ell = 1, m = 2$, (2) holds for all $\alpha \in [0,1]$ if

$$\begin{bmatrix} E_0 & S^\top \\ S & E_2 \end{bmatrix} \prec 0, \tag{3a}$$
$$2E_1 \preceq S^\top + S. \tag{3b}$$

The proof follows as pre-and-post multiplying (3a) by $\begin{bmatrix} \alpha I & (1-\alpha)I \end{bmatrix}$, it leads to

$$\alpha^2 E_0 + \alpha(1-\alpha)2E_1 + (1-\alpha)^2 E_2 \preceq \alpha^2 E_0 + \alpha(1-\alpha)(S^\top + S) + (1-\alpha)^2 E_2 \prec 0.$$

So the idea should follow, for any $\ell = 1, m \in \mathbb{Z}_+$:

Lemma 4: When $\ell = 1$, (2) holds for all $\alpha \in [0,1]$ if there exist matrices S_k, $k = 1, \ldots, m-1$, such that

$$\begin{bmatrix} E_0 & S_k^\top \\ S_k & E_m \end{bmatrix} \prec 0, \quad k = 1, \ldots, m-1, \tag{4a}$$
$$S_k = S_{m-k}, \quad k = 1, \ldots, m-1, \tag{4b}$$
$$(m-1)\binom{m}{k} E_k \preceq S_k^\top + S_k, \quad k = 1, \ldots, m-1. \tag{4c}$$

Proof: Fix $k \in \{1, \ldots, m-1\}$. Applying (4a) to k and $m-k$, and then using the equality constraint (4b), gives

$$2\left[\alpha^m E_0 + (1-\alpha)^m E_m\right] + \left[\alpha^{m-k}(1-\alpha)^k + \alpha^k(1-\alpha)^{m-k}\right](S_k^\top + S_k) \prec 0.$$

On the other hand, applying (4c) at k and $m-k$, respectively, and using (4b) again yields

$$\alpha^{m-k}(1-\alpha)^k(m-1)\binom{m}{k}E_k \preceq \alpha^{m-k}(1-\alpha)^k(S_k^\top + S_k),$$
$$\alpha^k(1-\alpha)^{m-k}(m-1)\binom{m}{m-k}E_{m-k} \preceq \alpha^k(1-\alpha)^{m-k}(S_k^\top + S_k),$$

where $S_k = S_{m-k}$ is used in the second inequality. Combining these three inequalities gives

$$2\left[\alpha^m E_0 + (1-\alpha)^m E_m\right] + (m-1)\binom{m}{k}\left[\alpha^{m-k}(1-\alpha)^k E_k + \alpha^k(1-\alpha)^{m-k}E_{m-k}\right] \prec 0.$$

Summing the last inequality for $k = 1, \ldots, m-1$ and dividing by 2 gives

$$(m-1)\left[\alpha^m E_0 + \sum_{k=1}^{m-1}\binom{m}{k}\alpha^{m-k}(1-\alpha)^k E_k + (1-\alpha)^m E_m\right] \prec 0.$$

Hence (2) holds for all $\alpha \in [0,1]$. ∎

2.3 What about multi-variable?

Lemma 4 relaxes the DP-LMI for $\ell = 1$. On the other hand, what about $\ell > 1$? Let's examine the case $\ell = 2, m = 2$:

$$\begin{aligned}
E(\alpha_1,\alpha_2) = &\binom{2}{0}\binom{2}{0}\alpha_1^2\alpha_2^2 E_{0,0} &&+\binom{2}{1}\binom{2}{0}\alpha_1(1-\alpha_1)\alpha_2^2 E_{1,0} + \binom{2}{2}\binom{2}{0}(1-\alpha_1)^2\alpha_2^2 E_{2,0}\\
&+\binom{2}{0}\binom{2}{1}\alpha_1^2\alpha_2(1-\alpha_2)E_{0,1} &&+\binom{2}{1}\binom{2}{1}\alpha_1(1-\alpha_1)\alpha_2(1-\alpha_2)E_{1,1} + \binom{2}{2}\binom{2}{1}(1-\alpha_1)^2\alpha_2(1-\alpha_2)E_{2,1}\\
&+\binom{2}{0}\binom{2}{2}\alpha_1^2(1-\alpha_2)^2 E_{0,2} &&+\binom{2}{1}\binom{2}{2}\alpha_1(1-\alpha_1)(1-\alpha_2)^2 E_{1,2} + \binom{2}{2}\binom{2}{2}(1-\alpha_1)^2(1-\alpha_2)^2 E_{2,2}.
\end{aligned}$$

Lemma 5: When $\ell = 2, m = 2$, (2) holds for all $\alpha_1, \alpha_2 \in [0,1]$ if

$$\begin{bmatrix} E_{0,0} & S_1^\top \\ S_1 & E_{0,2}\end{bmatrix} \prec 0, \quad 3\binom{2}{0}\binom{2}{1}E_{0,1} \preceq S_1^\top + S_1, \tag{5a}$$
$$\begin{bmatrix} E_{2,0} & S_2^\top \\ S_2 & E_{2,2}\end{bmatrix} \prec 0, \quad 3\binom{2}{2}\binom{2}{1}E_{2,1} \preceq S_2^\top + S_2, \tag{5b}$$
$$\begin{bmatrix} E_{0,0} & S_3^\top \\ S_3 & E_{2,0}\end{bmatrix} \prec 0, \quad 3\binom{2}{1}\binom{2}{0}E_{1,0} \preceq S_3^\top + S_3, \tag{5c}$$
$$\begin{bmatrix} E_{0,2} & S_4^\top \\ S_4 & E_{2,2}\end{bmatrix} \prec 0, \quad 3\binom{2}{1}\binom{2}{2}E_{1,2} \preceq S_4^\top + S_4, \tag{5d}$$
$$\begin{bmatrix} E_{0,0} & S_5^\top \\ S_5 & E_{2,2}\end{bmatrix} \prec 0, \quad \begin{bmatrix} E_{0,2} & S_6^\top \\ S_6 & E_{2,0}\end{bmatrix} \prec 0, \quad 3\binom{2}{1}\binom{2}{1}E_{1,1} \preceq S_5^\top + S_5 + S_6^\top + S_6. \tag{5e}$$

Proof: Applying the one-dimensional relaxation argument to the first four LMIs in (5) gives

$$\alpha_1^2 E_{0,0} + 3\binom{2}{0}\binom{2}{1}\alpha_1(1-\alpha_1)E_{0,1} + (1-\alpha_1)^2 E_{0,2} \prec 0,$$
$$\alpha_1^2 E_{2,0} + 3\binom{2}{2}\binom{2}{1}\alpha_1(1-\alpha_1)E_{2,1} + (1-\alpha_1)^2 E_{2,2} \prec 0,$$
$$\alpha_2^2 E_{0,0} + 3\binom{2}{1}\binom{2}{0}\alpha_2(1-\alpha_2)E_{1,0} + (1-\alpha_2)^2 E_{2,0} \prec 0,$$
$$\alpha_2^2 E_{0,2} + 3\binom{2}{1}\binom{2}{2}\alpha_2(1-\alpha_2)E_{1,2} + (1-\alpha_2)^2 E_{2,2} \prec 0.$$

Pre-and-post multiplying the two diagonal LMIs in (5e) by

$$\begin{bmatrix}\alpha_1\alpha_2 I & (1-\alpha_1)(1-\alpha_2)I\end{bmatrix}, \quad \begin{bmatrix}\alpha_1(1-\alpha_2)I & (1-\alpha_1)\alpha_2 I\end{bmatrix},$$

respectively, and using the last inequality in (5e), we have

$$\alpha_1^2\alpha_2^2 E_{0,0} + \alpha_1^2(1-\alpha_2)^2 E_{0,2} + (1-\alpha_1)^2\alpha_2^2 E_{2,0} + (1-\alpha_1)^2(1-\alpha_2)^2 E_{2,2}$$
$$+3\binom{2}{1}\binom{2}{1}\alpha_1(1-\alpha_1)\alpha_2(1-\alpha_2)E_{1,1} \prec 0.$$

Multiplying the four edge inequalities above by α_1^2, $(1-\alpha_1)^2$, α_2^2, and $(1-\alpha_2)^2$, respectively, and adding the diagonal inequality gives

$$3\sum_{i=0}^{2}\sum_{j=0}^{2}\binom{2}{i}\binom{2}{j}\alpha_1^{2-i}(1-\alpha_1)^i\alpha_2^{2-j}(1-\alpha_2)^j E_{i,j} = 3E(\alpha_1,\alpha_2) \prec 0.$$

Therefore, $E(\alpha_1,\alpha_2) \prec 0$ for all $\alpha_1,\alpha_2 \in [0,1]$. ■

Definition 2 (Index ordering function): For an ℓ-dimensional degree-m multi-index $\mathbf{i}$, define the radix-$(m+1)$ ordering function

$$\mathrm{ind}_m(\mathbf{i}) = \sum_{r=1}^{\ell} i_r(m+1)^{\ell-r}, \quad \mathbf{i} = (i_1,\ldots,i_\ell) \in \{0,\ldots,m\}^\ell.$$

Lemma 6: For the case $m=2$, suppose there exist matrices $S_{\mathbf{i},\mathbf{j}}$ such that

$$\begin{bmatrix} E_{\mathbf{i}} & S_{\mathbf{i},\mathbf{j}}^\top \\ S_{\mathbf{i},\mathbf{j}} & E_{\mathbf{j}}\end{bmatrix} \prec 0, \quad \forall \mathbf{i},\mathbf{j} \in \{0,2\}^\ell, \mathrm{ind}_2(\mathbf{i}) < \mathrm{ind}_2(\mathbf{j}), \tag{6a}$$

$$(2^\ell - 1)\left(\prod_{r=1}^{\ell}\binom{2}{k_r}\right) E_{\mathbf{k}} \preceq \sum_{\substack{\mathbf{i}+\mathbf{j}=2\mathbf{k}\\ \mathbf{i},\mathbf{j}\in\{0,2\}^\ell\\ \mathrm{ind}_2(\mathbf{i})<\mathrm{ind}_2(\mathbf{j})}} \left(S_{\mathbf{i},\mathbf{j}}^\top + S_{\mathbf{i},\mathbf{j}}\right), \quad \forall \mathbf{k} \in \{0,1,2\}^\ell \setminus \{0,2\}^\ell. \tag{6b}$$

Then

$$E(\boldsymbol{\alpha}) = \sum_{i_1=0}^{2}\cdots\sum_{i_\ell=0}^{2}\left[\prod_{r=1}^{\ell}\binom{2}{i_r}\alpha_r^{2-i_r}(1-\alpha_r)^{i_r}\right]E_{i_1,\ldots,i_\ell} \prec 0, \quad \forall \boldsymbol{\alpha} \in [0,1]^\ell.$$

2.4 A unified version of relaxation

Definition 3 (Interior indices covered by a vertex pair): For $\mathbf{i},\mathbf{j} \in \{0,m\}^\ell$, define

$$\kappa(a,b) = \begin{cases}\{a\}, & a=b,\\ \{1,\ldots,m-1\}, & a\neq b,\end{cases}$$

$$\mathcal{K}_{\mathbf{i},\mathbf{j}} = \kappa(i_1,j_1)\times\cdots\times\kappa(i_\ell,j_\ell), \qquad \mathcal{K}_{\mathbf{i},\mathbf{j}}^{+} = \{\mathbf{k}\in\mathcal{K}_{\mathbf{i},\mathbf{j}} : \mathrm{ind}_m(\mathbf{k}) \le \mathrm{ind}_m(\mathbf{i}+\mathbf{j}-\mathbf{k})\}.$$

Example 4: Let $m=3$, $\ell=3$, and take $\mathbf{i}=(0,0,0)$, $\mathbf{j}=(3,3,3)$. Then

$$\mathcal{K}_{\mathbf{i},\mathbf{j}} = \{1,2\}\times\{1,2\}\times\{1,2\},$$

i.e.,

$$\mathcal{K}_{\mathbf{i},\mathbf{j}} = \{(1,1,1),(1,1,2),(1,2,1),(1,2,2),$$
$$(2,1,1),(2,1,2),(2,2,1),(2,2,2)\}.$$

The mirror of $\mathbf{k}$ is $\mathbf{i}+\mathbf{j}-\mathbf{k} = (3,3,3)-\mathbf{k}$. Since ind_3 is the radix-4 ordering function,

$$\mathcal{K}_{\mathbf{i},\mathbf{j}}^{+} = \{(1,1,1),(1,1,2),(1,2,1),(1,2,2)\}.$$

The mirror relation gives the slack-variable identifications

$$S_{(0,0,0),(3,3,3)} = S_{(1,1,1),(2,2,2)} = S_{(1,1,2),(2,2,1)} = S_{(1,2,1),(2,1,2)} = S_{(1,2,2),(2,1,1)}.$$

That is:

$$S_{\mathbf{i},\mathbf{j}} = S_{\mathbf{k},\mathbf{i}+\mathbf{j}-\mathbf{k}} \quad \forall \mathbf{i},\mathbf{j}\in\mathcal{V}_m,\ \mathrm{ind}_m(\mathbf{i})<\mathrm{ind}_m(\mathbf{j}),\ \mathbf{k}\in\mathcal{K}_{\mathbf{i},\mathbf{j}}^{+}.$$

For general ℓ and m, we can have a unified version of relaxation:

Theorem 1: Let $\ell, m \in \mathbb{Z}_+$. Let

$$\mathcal{I}_m = \{0, \dots, m\}^\ell, \quad \mathcal{V}_m = \{0, m\}^\ell.$$

Suppose there exist matrices $S_{\mathbf{i},\mathbf{j}}$ satisfying

$$\begin{bmatrix} E_{\mathbf{i}} & S_{\mathbf{i},\mathbf{j}}^\top \\ S_{\mathbf{i},\mathbf{j}} & E_{\mathbf{j}} \end{bmatrix} \prec 0, \quad \forall \mathbf{i}, \mathbf{j} \in \mathcal{V}_m, \, \mathrm{ind}_m(\mathbf{i}) < \mathrm{ind}_m(\mathbf{j}), \tag{7a}$$

$$S_{\mathbf{i},\mathbf{j}} = S_{\mathbf{k},\mathbf{i}+\mathbf{j}-\mathbf{k}} \quad \forall \mathbf{i}, \mathbf{j} \in \mathcal{V}_m, \, \mathrm{ind}_m(\mathbf{i}) < \mathrm{ind}_m(\mathbf{j}), \, \mathbf{k} \in \mathcal{K}_{\mathbf{i},\mathbf{j}}^+, \tag{7b}$$

$$\begin{aligned}(m^\ell - 1)\left(\prod_{r=1}^{\ell} \binom{m}{k_r}\right) E_{\mathbf{k}} \preceq & \sum_{\substack{\mathbf{i},\mathbf{j} \in \mathcal{V}_m, \\ \mathrm{ind}_m(\mathbf{i}) < \mathrm{ind}_m(\mathbf{j}) \\ \mathbf{k} \in \mathcal{K}_{\mathbf{i},\mathbf{j}}^+}} \left(S_{\mathbf{k},\mathbf{i}+\mathbf{j}-\mathbf{k}}^\top + S_{\mathbf{k},\mathbf{i}+\mathbf{j}-\mathbf{k}}\right) \quad , \quad \forall \mathbf{k} \in \mathcal{I}_m \setminus \mathcal{V}_m. \\ & + \sum_{\substack{\mathbf{i},\mathbf{j} \in \mathcal{V}_m, \\ \mathrm{ind}_m(\mathbf{i}) < \mathrm{ind}_m(\mathbf{j}) \\ \mathbf{k} \in \mathcal{K}_{\mathbf{i},\mathbf{j}} \setminus \mathcal{K}_{\mathbf{i},\mathbf{j}}^+}} \left(S_{\mathbf{i}+\mathbf{j}-\mathbf{k},\mathbf{k}}^\top + S_{\mathbf{i}+\mathbf{j}-\mathbf{k},\mathbf{k}}\right)\end{aligned} \tag{7c}$$

Then $E(\boldsymbol{\alpha}) \prec 0, \forall \boldsymbol{\alpha} \in [0,1]^\ell$.

Remark 1: The coefficient $m^\ell - 1$ is the number of nontrivial vertex-pair directions and interior positions seen from any fixed vertex. Indeed, choose d coordinates along which the second vertex differs from the fixed one, and choose one of the $m-1$ interior Bernstein indices in each selected coordinate. This gives $\binom{\ell}{d}(m-1)^d$ terms for each $d = 1, \dots, \ell$, and

$$\sum_{d=1}^{\ell} \binom{\ell}{d} (m-1)^d = m^\ell - 1.$$

This is the multiplicity with which each vertex coefficient is generated when all vertex-pair relaxations are combined, so the same factor is used in (7c).

When $m = 2$, $m^\ell - 1 = 2^\ell - 1$, and $\mathbf{k} \in \mathcal{K}_{\mathbf{i},\mathbf{j}}$ is equivalent to $\mathbf{i} + \mathbf{j} = 2\mathbf{k}$. Therefore, Theorem 1 reduces to Lemma 6, with the slack variables identified by (7b). When $\ell = 1$, $m^\ell - 1 = m - 1$, and the only vertex pair is $(0, m)$. Then (7b) becomes $S_{0,m} = S_{k,m-k}$ for the ordered half $k \le m - k$, while (7c) uses $S_{m-k,k}$ for the mirrored half. This is the unified-version counterpart of the mirrored slack constraint $S_k = S_{m-k}$ in (4b). If the common endpoint slack $S_{0,m}$ is allowed to depend on k, this specialization recovers the less conservative one-dimensional formulation in Lemma 4.